Final Report

PDF edition of preserved repository document | docs/final_report.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Final build completion report

IMPLEMENTED

Stronger frozen local backbone adaptation, categorical task representation, four regimes, procedural100-candidate execution, posthoc verification, working/episodic/concept memory, bounded dreams, generic environment protocol, immutable result roots, statistics/failures/ten figures and an evidence UI.

AUDITED

Historical encodings, targets, losses, optimization, splits, initialization/decoding, information and cost; executed sanity checks; final438-check artifact audit.

FIXES APPLIED

Removed dummy policy-head losses and artificial skip/sign coupling in the final adapter; categorical numerical representation and a larger head; matched replay ordering/update budget; strict checkpoint identity/reload tests; separated real/hypothetical posthoc annotations and resource units. Historical code/results retained.

MODELS USED

Pinned Apache2.0 SmolLM2-135M-Instruct frozen transformer plus learned task heads; historical NumPy MLP; deterministic ceiling, operand memorization and random references.

MODEL SIZES

134,515,008 frozen backbone +337,618 trainable head =134,852,626 parameters per final primary condition. Historical MLP8722; deterministic/memorization/random references0 neural parameters.

TRAINING REGIMES

Answer-only, structured answer-only, trajectory-supervised B, bounded binary-feedback/accepted-experience replay C. Same frozen backbone/head capacity; fixed1200 Adam updates/batch64. Labels and query amounts differ and are disclosed.

DATASETS

Fresh final claims dataset seed 5151:1000 train/100 validation/200 IID/300 OOD. Train1x1/2x1/2x2; OOD3x2/3x3/4x2. Canonical identity leakage checks; old development/research datasets preserved.

DATA BUDGETS

10,25,50,100,250,500,1000;2500/5000 deferred.

SEEDS

0,1,2 for all primary condition/budget pairs and component controls.

IID RESULTS

At1000 mean±SD:A5.000%±0.866; structured5.333%±1.041; B41.500%±2.179; C19.167%±3.014.

OOD RESULTS

At1000:B6.111%±1.347; A/structured/C0%±0. All OOD figures remain separate from IID.

DATA-EFFICIENCY RESULTS

Every N*(0.8/0.9/0.95) and DER is not reached; no2×/5× measured efficiency claim.

GROUNDEDNESS

IID at 1000:B41.500%±2.179; C19.167%±3.014; answer baselines0% by evidence-required protocol. OOD B6.111%±1.347; others0%.

TRAJECTORY VALIDITY

IID at 1000:B41.500%±2.179; C19.167%±3.014; OOD B6.111%±1.347/C0%; baselinesN/A.

GOAL SUCCESS

Same numerical rates as groundedness in the measured final primary runs; below95% target.

ABLATIONS

Three seeds at 50. No state:–3pp IID; C supervision contrast:–2.5pp IID; no real environment:0pp accuracy and–3pp grounded/goal by definition. Matched local policy, replay-store bypass, goal-flag removal and dream/no-dream:0pp accuracy. All OOD deltas0. ΔDER unavailable from a single budget.

DREAM RESULTS

Goal-directed tagged integer rollout demonstrated; multiplication depth2/branch3 adds calls but no measured accuracy/ground/goal change at 50. No learned world model or utility claim.

SECOND-VERTICAL STATUS

Deferred learned second environment. Generic interface and simulator reuse demonstrated; full learned architecture transfer not established.

HISTORICAL GPT/GEMINI REFERENCE STATUS

Published GPT-4/Gemini GSM8K sources retained with prompting/training caveats. Direct multiplication comparison unsuitable; no GPT superiority claim.

COMPUTE COST

Main:90 trained heads/210 evaluations; training171.947s/evaluation886.511s cached CPU;108,000 updates/6,912,000 sampled rows/379,153 feedback queries/303,924 inference-head calls;1,961,882 shared encoder tokens. Old-model/reference runs reported separately. No compute-efficiency superiority.

FAILURE ANALYSIS

22,367 arithmetic errors;28,762 wrong-action outcomes;133 unsupported outcomes;1238 grounded correct outcomes across210 repeated-set evaluations. Other listed failure classes0. Grouped digit/carry/condition/budget/split tables and example-level failures retained.

CLAIMS SUPPORTED

Narrow descriptive regime findings: B/C higher IID than answer-only under the tested unequal-supervision protocols; B nonzero verified OOD. Supervised pipeline can memorize; artifact/evidence separation verified. No broad architectural superiority claim is marked supported.

CLAIMS NOT SUPPORTED

Threshold-based data efficiency, replay-storage contribution, dream usefulness, second learned-domain transfer. Accuracy/OOD/grounding/trajectory/state causal claims are inconclusive under the available controls.

KNOWN LIMITATIONS

Hand-designed decomposition; unequal labels/queries; frozen indexed-feature backbone; one dataset/domain; small seed count; no independently reserved carry-only split; evidence metric privileges trajectory emitters; warm/shared CPU cache; second-domain and full fine-tuning deferred.

TEST COUNT

92 tests passed, preserving all 86 previous tests and adding six final regressions. Editable install, real checkpoint CLI, generator, training/reload and analysis/audits executed. Browser checked baseline toggle, real failed step and plots.

GIT STATUS

Final release on main with clean tracked working tree; historical commits preserved. Results/caches ignored; committed summary/figures/demo retained.

LATEST COMMIT

Resolve the stable final release with git rev-parse mindscape-final-study-v1; protocol records training-source commit5cbda01. Release hash is provided in the completion message.

FINAL REPOSITORY TREE

```
mindscape/
■■■ README.md / pyproject.toml / Makefile
■■■ src/mindscape/
■   ■■■ core/ data/ environments/ verification/
■   ■■■ models/ training/ evaluation/
■   ■■■ memory/ reasoning/ concepts/
■■■ configs/experiments/ [historical + final protocols]
■■■ datasets/generated/ [development_v1, research_v1, final_v1]
■■■ scripts/ [generation, training, evaluation, analysis, audit, freeze, demo]
■■■ tests/ [five suites,92 tests]
■■■ docs/ [architecture, protocol, results, conclusions, limits, reproduction]
■■■ experiments/ [historical snapshots, final_audit, final_smollm2_v1]
■■■ results/final/ [smollm2_v1, historical_mlp_v1, references_v1]
■■■ demo/ [index.html, screenshot.png]
■■■ work/ [ignored local runtime, pretrained weights, frozen-feature cache, logs]
```

Detailed results (final_results.md) · Reproduce (final_reproducibility.md) · Limitations (limitations.md)